

Practice Quiz 10A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 10

Images and inverse images; the geometric sum formula.

1. (a) Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$, and define

$$f(1) = a, \quad f(2) = b, \quad f(3) = b, \quad f(4) = c.$$

For $X = \{1, 3\}$, compute $f(X)$ and $f^{-1}(f(X))$. Are the latter and X equal?

Solution $f(X) = \{a, b\}$. The elements mapped into $\{a, b\}$ are 1, 2, 3, so

$$f^{-1}(f(X)) = \{1, 2, 3\} \neq \{1, 3\} = X.$$

The extra element 2 appears because $f(2) = f(3) = b$. ■

- (b) State the geometric sum formula for $1 + r + \cdots + r^n$, including its hypotheses, and prove it by induction on n .

Solution For every real $r \neq 1$ and integer $n \geq 0$,

$$\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1}.$$

For $n = 0$, both sides equal 1. Suppose the formula holds for n . Then

$$\begin{aligned} \sum_{i=0}^{n+1} r^i &= \frac{r^{n+1} - 1}{r - 1} + r^{n+1} \\ &= \frac{r^{n+1} - 1 + r^{n+1}(r - 1)}{r - 1} = \frac{r^{n+2} - 1}{r - 1}. \end{aligned}$$

This is the formula for $n + 1$, so induction proves the result. ■